# CATH PLAN · Case 5

66M, 74 kg · scheduled 13:00 · PCI to left main bifurcation and mid-RCA with Impella CP planned (ischemic cardiomyopathy) · pathway: known intervention

**Synthetic teaching content. Not clinical advice. Not a medical device. DRAFT for the chair to rule on.**

**Thresholds: PROPOSED values, not yet verified by the chair. Template logic (likelihood weights and bands, prep rules, Imaging triggers, Heart Team factors, stage assignment, Labs' eGFR cutoff) is also unreviewed.**

| 6 | 2.0 | 3 / 4 | 1 |
|---|---|---|---|
| flags from 4 agents | contrast ratio (published cutoff 3.7, Laskey) | angio findings flagged (calcium, flow, bifurcation, multivessel) | devices out of stock |

Contrast ratio = planned contrast (mL) ÷ eGFR. The published ratios use creatinine clearance, not eGFR (docs/EVIDENCE.md E1, E2): risk rose from 2 (Gurm 2011) and 3.7 was a discriminating cutoff (Laskey 2007). No guideline sets a limit; Gurm: risk rose from 2 and was clearest at 3 or more (EVIDENCE E1 does not record an advised limit); Laskey 3.7 is a discriminating cutoff, not a threshold for harm. The chair confirms the basis.

## Lesion map (Imaging)

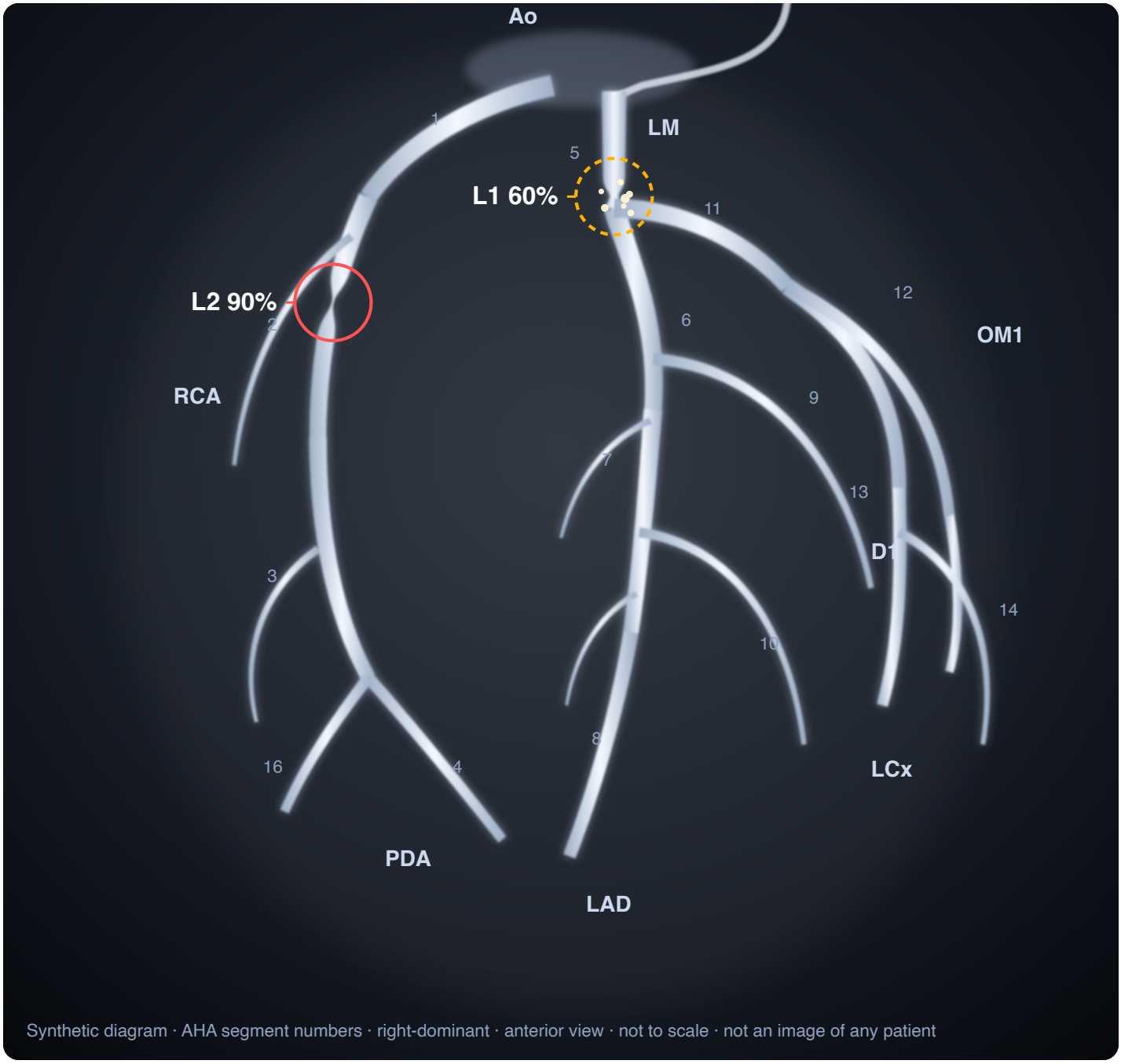
| Lesion | Vessel | Stenosis | TIMI | Calcium | Flag |
|---|---|---|---|---|---|
| L1 | LM bifurcation | 60% | TIMI 3 | calcium moderate | bifurcation Medina 1,0,1 |
| L2 | mid-RCA | 90% | TIMI 3 | calcium none | none |

Triggers: calcium moderate or severe; bifurcation, ostial or left main; more than one lesion (multivessel). · Cath report: indication ischemic cardiomyopathy with angina; access femoral and radial; hemodynamics not recorded

## Rules that fired

- Contrast ratio 2.0 is at or above 2.0, where risk began to rise in the Gurm 2011 registry (E1; creatinine clearance, not eGFR). Chair to decide.
- Heart Team factors present (2021 ACC/AHA/SCAI; E7, E8): left main disease. SYNTAX score is not computed here. These are packet factors; the chair decides what follows.

## Angiographic findings (synthetic)




**L1 · LM bifurcation** (AHA segment 5)
60% stenosis · TIMI 3 (normal filling) · calcium moderate · flag bifurcation Medina 1,0,1

**L2 · mid-RCA** (AHA segment 2)
90% stenosis · TIMI 3 (normal filling) · calcium none · flag none

Drawn from the angio lines in the packet by tools/angio.py: right-dominant tree, anterior view, AHA segment numbers, lumen narrowed to the stated percentage, calcium shown as bright specks, distal filling dimmed when TIMI flow is below 3. Solid red ring = stenosis of 70% or more; dashed amber ring = under 70%, or a CCTA estimate not yet confirmed. A Medina 1,1,1 flag is drawn as a narrowing of about half the proximal LAD and LCx. A teaching diagram, not an image of any patient and not to scale.

## Equipment manifest

| Item (from the packet) | Stock |
|---|---|
| Impella CP | **OUT: 0 on hand · no resupply scheduled** |
| 7F guide | 2 on hand |
| support wire | 4 on hand |
| NC balloon 3.5 mm | 2 on hand |
| DES 3.5x28 mm | 2 on hand |
| standard PCI tray | 4 on hand |

## Prep check (devices against findings)

| | |
|---|---|
| **Scope** | Stage the devices named in the packet |
| **Stock** | **OUT: Impella CP** |

| On the shelf, by class (from the STOCK line) | |
|---|---|
| Guide catheters | 6F guide 4, 7F guide 2 |
| Diagnostic catheters | 5F diagnostic catheters 8 |
| Wires | support wire 4 |
| Balloons | NC balloon 3.0 mm 2, NC balloon 3.5 mm 2 |
| Stents (DES) | DES 3.5x24 mm 3, DES 3.5x28 mm 2, DES 3.0x18 mm 0 (no resupply scheduled) |
| Intravascular imaging | IVUS catheter 1 |
| Support devices | Impella CP 0 (no resupply scheduled) |

**Questions for the chair** (the plan asks, it does not choose a device):

- 2 lesions on angiography and 1 stent named in the packet: chair to confirm what to stage.
- Calcium moderate on L1 LM bifurcation (angiography): the packet names no calcium-modification device or intravascular imaging catheter. Chair to decide whether any is needed.
- Left main or bifurcation on L1 (angiography): the packet names no intravascular imaging catheter. Chair to decide.
- EF 28% in the packet. The 2015 SCAI/ACC consensus names high-risk PCI (multivessel, left main, last conduit) with severely reduced EF but gives no EF cutoff (E10). Support devices on the shelf: Impella CP 0. Chair to decide whether support is needed.

## Flags from the agents

**Imaging**
- calcium moderate · L1 LM bifurcation 60% · ref: ACC/AHA 2021 LM
- bifurcation · L1 LM bifurcation 60% · ref: ACC/AHA 2021 LM
- multivessel · L1 LM bifurcation 60% · L2 mid-RCA 90%

**Labs**
- Cr 1.4 vs 1.1 yesterday · ref: KDIGO AKI
- eGFR 54 vs est. contrast 110 mL (calc: 110 mL / eGFR 54 = 2.0) · ref: Gurm 2011 ratio

**Meds**
- ✅ nothing flagged

**Equipment**
- Impella CP 0 on hand · plan: "Impella CP planned" · STOCK: "no resupply scheduled"

Relay: Imaging → Equipment, Labs → Meds.

## Summary and approvals

| | |
|---|---|
| **Ruling recorded** | none yet |
| **Flags** | Imaging 3 · Labs 2 · Meds 0 · Equipment 1 |
| **Rules that fired** | Contrast ratio 2.0 is at or above 2.0, where risk began to rise in the Gurm 2011 registry (E1; creatinine clearance, not eGFR). Chair to decide. Heart Team factors present (2021 ACC/AHA/SCAI; E7, E8): left main disease. SYNTAX score is not computed here. These are packet factors; the chair decides what follows. |

**For the chair to decide / check** (the agents only flag):

- Thresholds in thresholds.json are proposed: the chair verifies them before this plan is used.
- Template logic (likelihood weights and bands, prep rules, Imaging triggers, Heart Team factors, stage assignment, Labs' eGFR cutoff) is unreviewed.

**Approvals**

No approval decisions recorded yet.

For the chair's use: print and mark.

- [ ] Approve as drafted
- [ ] Chair changes the plan (record the change)
- [ ] Equipment issue noted

**Status**

This draft is not approved until the chair presses Allow once on the Switch card.

AWAITING CHAIR

Sources: packet (day2.md) · briefs from Imaging, Labs, Meds, Equipment · thresholds.json · evidence table docs/EVIDENCE.md (E1 to E16, mostly secondary summaries, chair verifies). Generated 2026-11-05 19:10 by plan.py (template, no model).
Provenance: template a0c1d013d2 · thresholds 51ec501c23 · plan.py d0e4e2d2e3 · packet id n/a. "ref:" on a flag names a published criterion; it is not an instruction, and the criteria are unverified until the chair has checked them against the sources (docs/EVIDENCE.md).